



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION - SEMESTER I (2020/2021)
CSC1041 - PROGRAMMING LABORATORY I

Answer **ALL** Questions

Time Allowed: **Three Hours**

Save your C programs as S19XXXQ1.c and S19XXXQ2.c where S19XXX is your registration number. Extra marks will be awarded for using good programming practices.

1. [50 Marks]

Create a C program to order pizza from a pizza store. The program should collect user data including the customer name, nic, mobile, pizza types, topping details, and quantities. A customer can buy a maximum of 10 pizzas at a time. Available Pizza types and relevant prices for the regular size pizzas are as follows. For each Large size pizza an extra 250 rupees per item must be paid at the checkout.

Pizza Type	Hot & Spicy	Cheesy	Veggie	Meat	BBQ Chicken
Price (Rs.) [Regular Size]	500.00	650.00	500.00	700.00	750.00

Topping type	Pepperoni	Mushroom	Extra cheese	Onion	Green pepper
Price (Rs.)	100.00	150.00	200.00	150.00	100.00

A recurring menu should be implemented in order to place orders for different types of pizza. There should be separate functions to print the pizza price list and the toppings price list. After placing an order, a receipt should be printed as below with the purchase summary;

Example output:

-----Order Details-----

Customer name : Kamal
Customer ID : 12345678912v
Customer Phone : 1234566666

Regular	Hot & Spicy	- Onion	3 →	1950.00
Large	BBQ Chicken	- Pepperoni	1 →	1100.00
Large	Meat	- Mushroom	4 →	4400.00
TOTAL →				7450.00

-----Thank you-----

The expected menu functionality should be similar to the below example.

```
Example:
-----Pizza Store Menu-----
    1.Order Pizza.
    2.View Pizza Menu.
    3.View Toppings Menu.
    9.Exit.
Your Selection: 1
insert national id number : 1234567v
insert name: Kamal
insert mobile number: 0771233456
-----Pizza Prices-----
1. hot & spicy      500.00
2. cheesy           650.00
3. veggie           500.00
4. Meat             700.00
5. BBQ Chicken      750.00
-----
Select your choice: 3
-----Topping Prices-----
1. Pepperoni        100.00
2. Mushroom         150.00
3. Extra cheese     200.00
4. Onion            150.00
5. Green pepper     100.00
-----
Select your choice: 2
input pizza quantity: 2
-----select pizza size-----
1.Regular
2.Large
Select your choice: 2
```

2. [50 Marks]

It is required to create a Senior Citizen Registration System containing details of a group of Senior Citizens. For health and insurance purposes the BMI for each person should also be stored along with other personal details. Body mass index (BMI) is an inexpensive and easy screening method for understanding the weight category of a particular person. Create a C program to store personal details of these Senior Citizens, one person at a time. The program should have a recurring menu and ask for details below when the user selects “Add a Senior Citizen”.

Name, NIC, age, gender, height, weight, blood_group, BMI, and BMI_status.

There should be a separate function to calculate BMI using the below formula.

$$BMI = \frac{Weight}{Height * Height}$$

BMI status should be assigned according to their calculated BMI value as follows.

BMI Value	< 18.5	18.5-24.9	25-29.9	30-34.9	35<
status	Underweight	Normal	overweight	Obese	Extremely Obese

Create a separate function to view all the added Senior Citizens with their details.

Create a separate function to view obese elders if their BMI_status equals to *Obese* or *Extremely Obese*.

Example Main Menu:

Senior Citizen Registration System.

***** select an option from the menu. *****

1.Add a Senior Citizen.

2.View All Elders.

3.View Obese Elders.

9.Exit.

option:

View All Senior Citizens Example output:

```
-----Senior Citizens Details-----
Name      Age      BloodGroup  BMI-Status
Jhon      78        O+         Overweight
Charles   67        B-         Obese
Jennie    56        AB+        Normal
-----End of List-----
```